01 DECISION SNAPSHOT

M1 rendered evidence

Prepared for Wei He | 17 October 2026 | Synthetic PV investigation world

M1 PASS - REQUIRED EVIDENCE AND REVIEW PRESENT

408	10.52 s	6
Passed cases in supplied JUnit	Complete pilot duration	Original M1 RGB frames in batch

Complete pilot - source hashes match at report build

The verified recording contains 5,261 actual MuJoCo/PV/thermal states through tick 5,260. Seed 11 finishes its recorded action sequence and verification tail. The batch and review gates are stated below. [S1-S4]

The supplied test report records 408 passed, 0 failed, 0 errors and 0 skipped cases. Counts come from that single JUnit file; overlapping component and reviewer selections are not added. [S5]

Complete development pilots supplied: 3. Captured M1 RGB frames in the primary pilot: 2.

Scope of this report

Established by recorded inputs	Established by desktop/review inputs
Finite actual motion and instrument timing; explicit surrogate output and heat accounting.	Verified actual camera frames and three successful recorded pilot/replay pairs.
Recorded actions replayed without calling a policy; original recorded wall-clock cost retained.	Supplied independent M1 PASS; Wei's authorization is still required before M2.

This is a private evaluator report under trusted-policy/local. It makes no physical validation, diagnosis-performance or hostile-code isolation claim.

02 EVIDENCE AND READINESS

What passed, what remains

Current report inputs are explicit; historical checkpoints remain labelled.

Check	Evidence state	Interpretation
Supplied automated suite	408 / 408 passed	JUnit file; current-code scope depends on its own run provenance.
Primary action replay	Passed	Matched source recording; exact tick checks plus private state and runtime metrics.
Independent review	M1 PASS	Review verdict applies only to its stated scope.
Original M1 RGB frames	6	No image can be inferred from numeric state or a scientific geometry plot.
Desktop batch	Three completed pairs	Three seed-specific pilot/replay pairs are required for the planned gate.
Machine batch audit	Passed	Bound to this batch report by SHA-256; every recorded machine check must pass.
Full milestone gate	Passed	M2 remains outside this report's authorization.

Recorded action chronology

Attempt at (s)	Action ID	Recorded request
0.020	1	capture_rgb
0.332	2	move_camera
3.572	3	probe_voltage
7.020	4	set_voltage
7.620	5	wait
8.320	6	finish

All 22 render cases are included in the supplied executed suite. The action table lists requests; completed feedback and delivered evidence are assessed separately.

03 ELECTRICAL AND THERMAL STATE

Energy through the complete pilot

Saved state at each 2 ms tick; plotting does not rerun the simulator. [S2-S4]

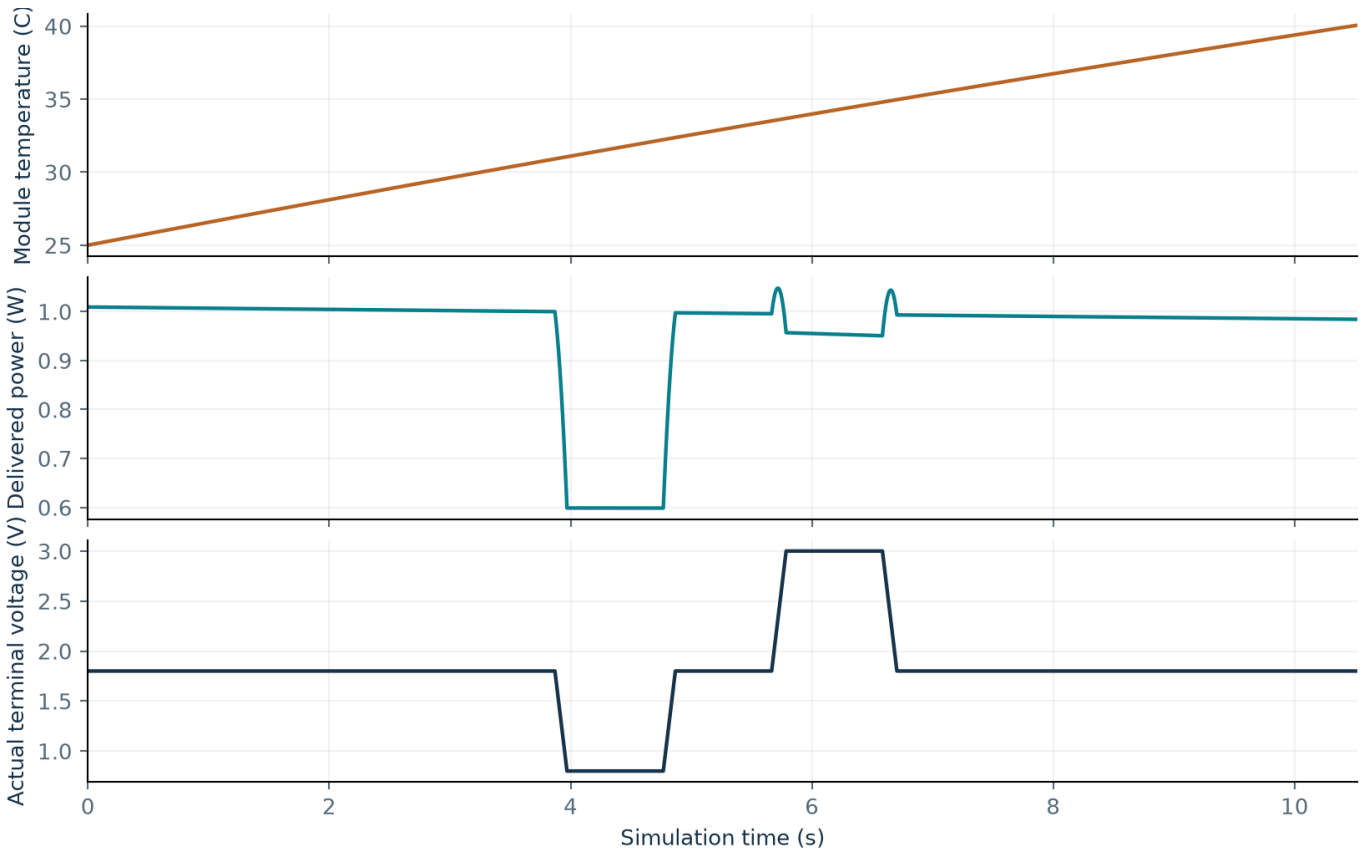


Figure 1. Temperature rises from 25.00 to 40.07 C while load/probe operations change delivered power. These are synthetic model states, not calibrated hardware measurements.

Recorded energy account	Joules
Absorbed incident energy	511.272000
Delivered electrical output	10.105492
Integrated heat loss	49.213309
Thermal storage change	451.953198

Residual = absorbed - output - heat loss - storage = -4.883e-11 J. Output integration freezes each tick's electrical input; the thermal step integrates linear cooling exactly. Plotted electrical values are end-of-tick samples, so integrating the plotted curve is not the ledger rule.

Across 5,261 true samples, max |P - V x I| = 0.000e+00 W.

04 ACTUAL MOTION

Servos follow finite trajectories

Saved MuJoCo positions; dashed curves are saved finite servo setpoints. [S2]

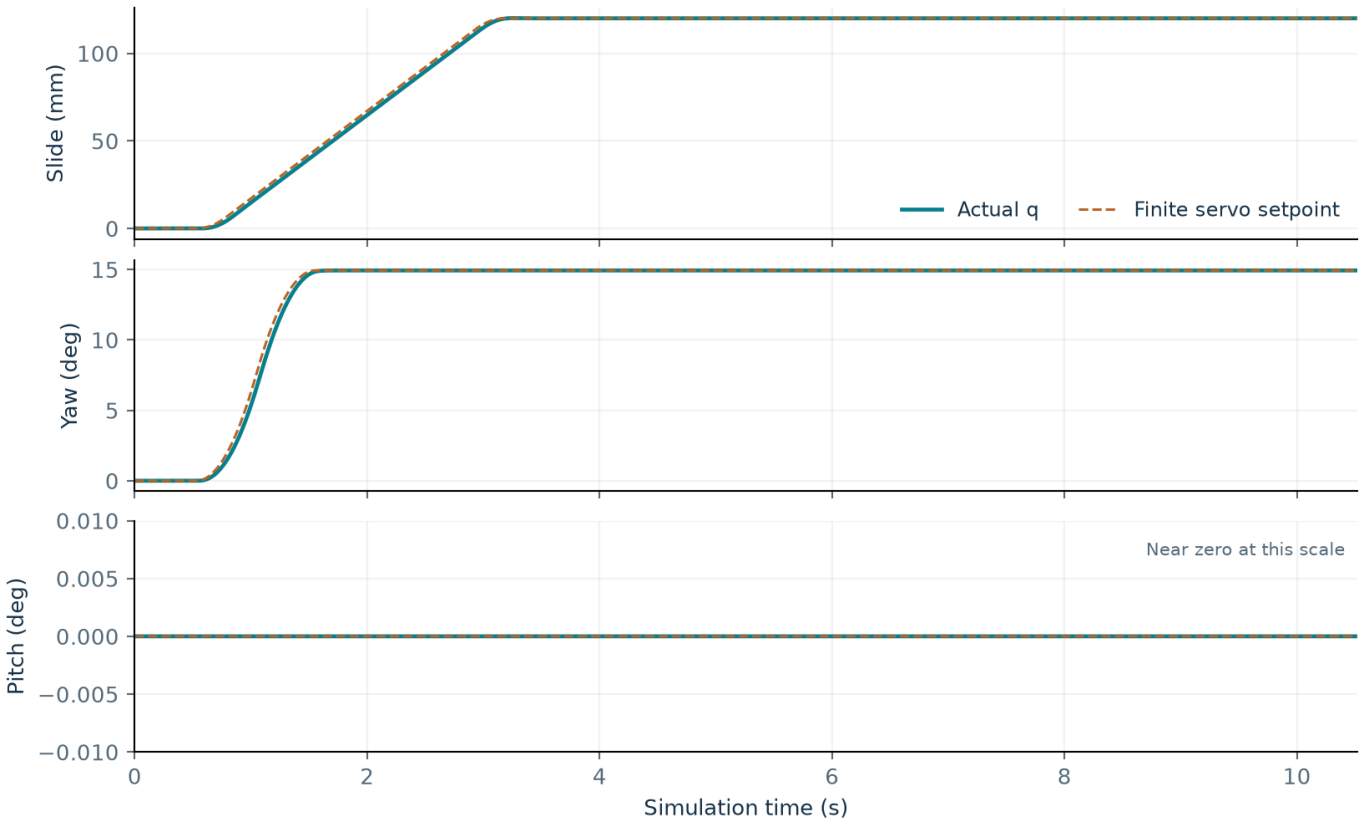


Figure 2. Slide, yaw and pitch evolve over simulation time. A setpoint curve is a command trajectory, while the solid curve is the achieved joint coordinate. Endpoint agreement alone does not establish settlement throughout every action.

Coordinate	Final actual	Final setpoint	Absolute error
Slide (mm)	120.000000	120.000000	0.000e+00
Yaw (deg)	14.931417	14.931417	2.862e-14
Pitch (deg)	-0.000000	0.000000	4.778e-95

Recorded travel: slide 0.120219 m, yaw 0.260602 rad, pitch 0.000000 rad. The actuator-work proxy is 0.000433337 J. It is reported separately and is not deducted from PV output.

Largest saved slide rate: 0.051243 m/s; largest saved setpoint rate: 0.050000 m/s. The trajectory rate cap applies to the commanded setpoint. Finite servo dynamics can briefly exceed that rate; the cap is not an exact bound on actual MuJoCo velocity.

05 SCENE AND CAMERA

Recorded camera path in 3D

Scientific geometry figure from configuration and actual pose records. [S2, S4]

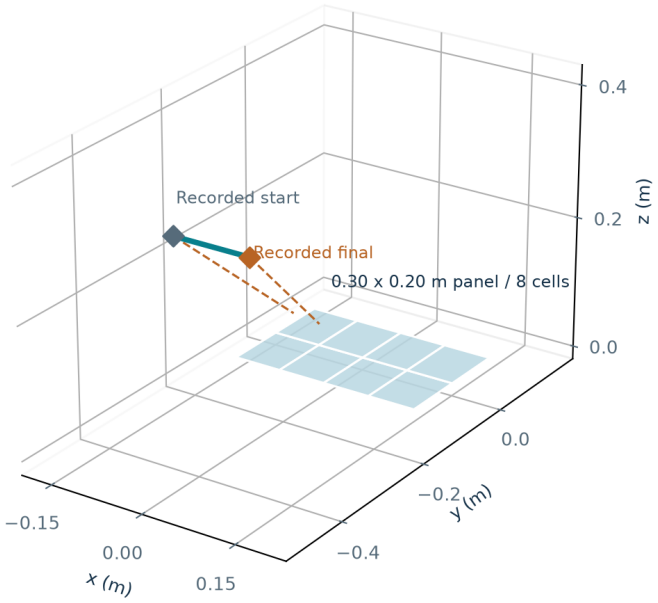


Figure 3. Panel geometry comes from the recorded configuration; camera centres and orientation rays come from actual stored transforms. The drawing omits the rig housing, lighting and occluders. It is not a MuJoCo RGB image or a visibility test.

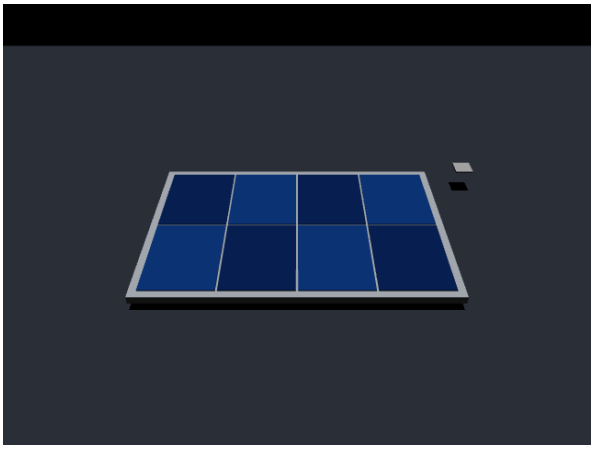
Seed / view	Capture / delivery (s)	Actual camera centre x, y, z (m)	Yaw (rad)
11 / Home	0.226 / 0.332	0.000000, -0.450000, 0.350000	0.000000
11 / Side	3.466 / 3.572	0.119996, -0.450000, 0.350000	0.260602
23 / Home	0.226 / 0.332	0.000000, -0.450000, 0.350000	0.000000
23 / Side	3.466 / 3.572	0.119996, -0.450000, 0.350000	0.260602
47 / Home	0.226 / 0.332	0.000000, -0.450000, 0.350000	0.000000
47 / Side	3.466 / 3.572	0.119996, -0.450000, 0.350000	0.260602

Poses above are saved at the exposure midpoint, not the final action endpoint. Each recorded exposure lasts 12 ms under the midpoint-snapshot approximation; camera and sensor delivery remain distinct from capture time. Actual RGB follows.

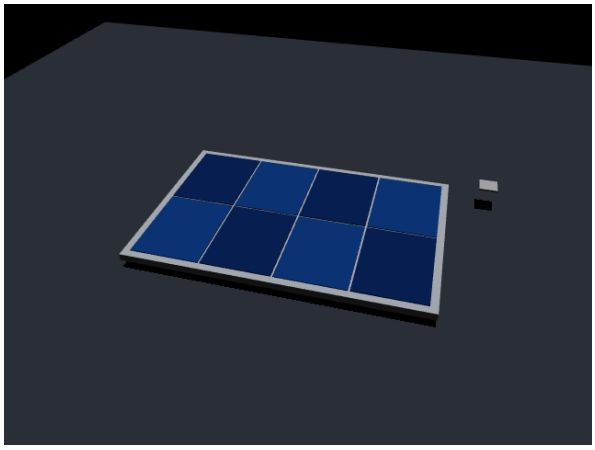
06 DESKTOP CAMERA EVIDENCE

Home and side views: all three seeds

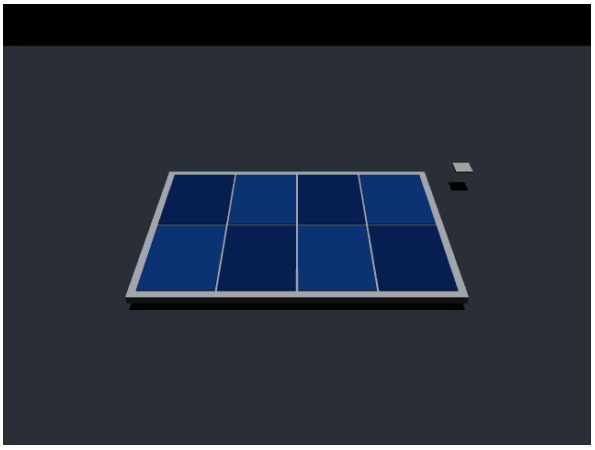
Six original PNGs; scaled for page layout without pixel editing or overlays.



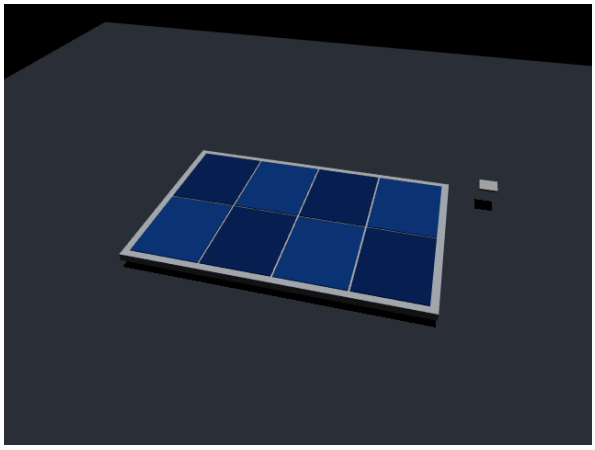
Seed 11 / Home / frame_000001
Capture 0.226 s; delivered 0.332 s.



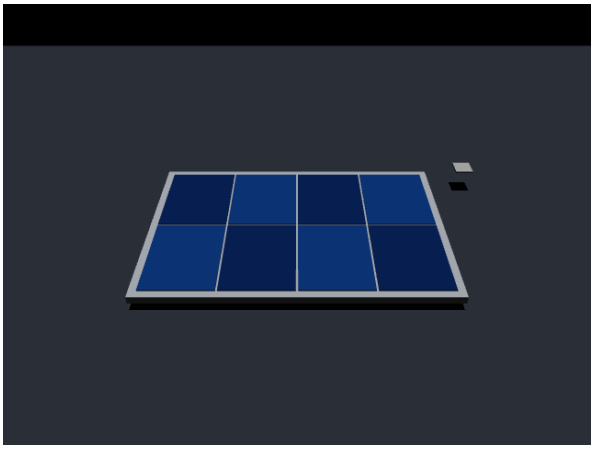
Seed 11 / Side / frame_000002
Capture 3.466 s; delivered 3.572 s.



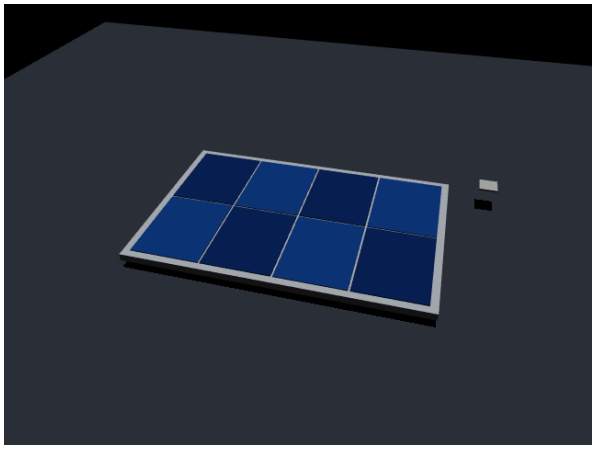
Seed 23 / Home / frame_000001
Capture 0.226 s; delivered 0.332 s.



Seed 23 / Side / frame_000002
Capture 3.466 s; delivered 3.572 s.



Seed 47 / Home / frame_000001
Capture 0.226 s; delivered 0.332 s.



Seed 47 / Side / frame_000002
Capture 3.466 s; delivered 3.572 s.

07 MEASURED CHOICE AND REPLAY

A tested load, then fresh verification

Public delivered readings support the choice; private state supports evaluation.

Probe V	Measured I (A)	V x I (W)	Capture / delivery (s)
0.800	0.748196	0.598557	4.600 / 4.620
1.800	0.553316	0.995969	5.500 / 5.520
3.000	0.317347	0.952041	6.400 / 6.420

Seed 11: the selected 1.8 V was one of the measured probe points. The completed finish action confirms a newer paired reading: 1.800 V x 0.546787 A = 0.984217 W, captured at 10.500 s and delivered at 10.520 s. Verification tail: 8.520 to 10.520 s. This is a measured choice among tested loads, not proof of the global optimum.

Development pilot / replay pairs

Seed	Pair gate	Sim. s	Truth rows	RGB pairs	Max pixel error
11	Passed	10.52	5261	2	0
23	Passed	10.52	5261	2	0
47	Passed	10.52	5261	2	0

Maximum private truth/metric difference across supplied replays: 0. Exact action ticks, observations, feedback, capture/delivery times and saved decisions are replayed and compared without querying the policy again. Numeric tolerances are 1e-10 absolute/relative; image limits are max 2 and mean 0.1 uint8 levels. These are same-pinned-host results, not a cross-platform guarantee.

Final private state at 10.520 s: 1.800 V, 0.546761 A, 0.984171 W. This later state is distinct from the delivered verification sample above.

What this establishes

The static synthetic scene now has actual rendered home/side evidence, finite camera motion, delivered probe readings, a fresh measured verification point and a completed tail for every development seed. The repeated deterministic scene does not establish causal diagnosis accuracy, hardware fidelity or broad scenario coverage. Reported policy wall time is local measured bookkeeping, not an inference benchmark. M2 still requires a separate user decision.

08 PROVENANCE AND REVIEW

A traceable checkpoint

Source snapshots, figure measurements and hashes accompany this PDF.

Independent review

Supplied verdict: **PASS_M1**. The review file is preserved without edits in the source snapshots. Its verdict is not promoted beyond the scope stated in that file. [S6] Wei executed the desktop suite, pilots and rendered replays. The independent review checked those retained artifacts and added separate non-GL tests; overlapping counts are not summed.

Matched replay report: passed; 5,261 private truth records and 49 numeric metric values compared. Maximum private metric difference: 0.0. [S7] Original recorded policy wall time: 0.02 s. Replay reuses decisions and does not time a new policy call.

Integrity and version scope

Manifest and every listed public/private artifact passed SHA-256 verification before plotting. The source checkpoint was completed at 2026-10-17T19:21:24.901706+00:00. At this report build, 0 source/configuration paths differ from that checkpoint. Historical traces remain valid evidence of their recorded version; they do not establish successful execution of later code.

Source	Retained input	SHA-256 prefix
S1	Verified source recording manifest	ab7713aba74db470
S2	Private actual-state trace	db21505bd7f93f47
S3	Recorded energy and cost metrics	a74e7bb748fc0fde
S4	Expanded private configuration	fef54964e97a8c39
S5	JUnit test evidence	a5081745d18fb3d0
S6	Independent review input	9ecf5f676029dc41
S7	Matched recorded-action replay report	1825ba7a61a76765
S10	Desktop batch execution report	cb6dbee1262f6d08
S11	Desktop machine evidence audit	3625125ef12997b8

Reproduction boundary

The report generator reads artifacts only. It opens no graphics context, executes no policy and runs no new episode. Full hashes, input locations, preserved snapshots, figure hashes, measurements, runtime package versions and the generation command are in the adjacent provenance JSON. No M2 authorization is inferred from this report.